observable averages diverge and we do not recover correct sampling in the limit of zero stepsize.

The symmetric schemes considered in the previous section are consistent however, each with their own limiting schemes corresponding to taking the limit of infinite friction in their update equations. As the momenta are continually redrawn from a normal distribution, we focus on the evolution of the configurations themselves. For the method denoted ⟦OBABO⟧, we have the limiting scheme

$$\begin{aligned}
\boldsymbol{p}_{n+1/2} &= \sqrt{k_{\mathrm{B}}\mathrm{T}}\boldsymbol{M}^{1/2}\mathbf{R}_n,\\
\hat{\boldsymbol{p}}_{n+1/2} &= \boldsymbol{p}_{n+1/2} - (h/2)\nabla U(\boldsymbol{q}_n),\\
\boldsymbol{q}_{n+1} &= \boldsymbol{q}_n + h\boldsymbol{M}^{-1}\hat{\boldsymbol{p}}_{n+1/2},\\
\hat{\boldsymbol{p}}_{n+1} &= \hat{\boldsymbol{p}}_{n+1/2} - (h/2)\nabla U(\boldsymbol{q}_{n+1}),\\
\boldsymbol{p}_{n+1} &= \sqrt{k_{\mathrm{B}}\mathrm{T}}\boldsymbol{M}^{1/2}\mathbf{R}_{n+1/2},
\end{aligned}$$

where $\mathbf{R}_n$ and $\mathbf{R}_{n+1/2}$ are independent normal random vectors with mean zero and unit variance. It is clear that if our only desire is to make use of the position data $\boldsymbol{q}$, then the final two steps are redundant in this algorithm, as we update the momentum vector only to redraw it at the beginning of the next iteration. Any information about the initial momentum vector $\boldsymbol{p}_n$ is unused in the scheme, making it simple to rewrite solely in terms of the position vector, giving us

$$\boldsymbol{q}_{n+1} = \boldsymbol{q}_n - \frac{h^2}{2}\boldsymbol{M}^{-1}\nabla U(\boldsymbol{q}_n) + h\sqrt{k_{\mathrm{B}}\mathrm{T}}\boldsymbol{M}^{-1/2}\mathbf{R}_n. \tag{7.38}$$

Denoting $\tau = h^2/2$, the update scheme (7.38) becomes the Euler-Maruyama scheme

$$\boldsymbol{q}_{n+1} = \boldsymbol{q}_n - \tau\boldsymbol{M}^{-1}\nabla U(\boldsymbol{q}_n) + \sqrt{2\tau k_{\mathrm{B}}\mathrm{T}}\boldsymbol{M}^{-1/2}\mathbf{R}_n, \tag{7.39}$$

which is known to give an error in long-time averages $\mathcal{O}(\tau) = \mathcal{O}(h^2)$, just as our analysis suggests.

If we instead consider the ⟦BAOAB⟧ scheme we find that unlike the ⟦OBABO⟧ scheme we no longer have redundant steps:

$$\begin{aligned}
\boldsymbol{p}_{n+1/2} &= \boldsymbol{p}_n - (h/2)\nabla U(\boldsymbol{q}_n),\\
\boldsymbol{q}_{n+1/2} &= \boldsymbol{q}_n + (h/2)\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1/2},\\
\hat{\boldsymbol{p}}_{n+1/2} &= \sqrt{k_{\mathrm{B}}\mathrm{T}}\boldsymbol{M}^{1/2}\mathbf{R}_n,\\
\boldsymbol{q}_{n+1} &= \boldsymbol{q}_{n+1/2} + (h/2)\boldsymbol{M}^{-1}\hat{\boldsymbol{p}}_{n+1/2},\\
\boldsymbol{p}_{n+1} &= \hat{\boldsymbol{p}}_{n+1/2} - (h/2)\nabla U(\boldsymbol{q}_{n+1}),
\end{aligned}$$